

**Investigation of Trans-Saccadic Information
Transfer and Varying Spatial Attention
Mechanisms in Goal-Directed Visual Search**

Anna Rose Hunt-Isaak

Supervisor: Dr. Suresh Krishna

1. Introduction

As humans view the world, they dynamically adjust their gaze to bring light from different scene locations onto their fovea – the portion of the retina which admits the highest visual acuity; this is typically achieved through rapid eye movements known as saccades. A long-standing question in the field of visual perception is to what extent (and through what mechanisms) visual operations integrate visual information from different fixations.

Early explorations of change blindness – a phenomenon in which significant changes to a visual scene go unnoticed – suggested that the effects of such information transfer across fixations are quite limited (Henderson & Hollingworth, 2003; Simons & Levin, 1997). It has even been proposed that visual perception and decision-making regarding subsequent eye movements occur completely independently at each new fixation, with no influence from prior fixations (Findlay, et al., 2001).

However, other research has challenged the purported compartmentalization of visual information of these findings. Various studies have demonstrated that both short and long-term visual memory accumulates across fixations and are accessible during scene viewing (Hollingworth, 2005; Varakin & Levin, 2006), including the use of spatial representations to plan fixation sequences and patterns of hand-eye coordination (Hayhoe, et al., 2003). It has been noted that, across saccades, one can access abstract featural and positional information of up to ~5 objects via this short-term visual memory (Irwin, 1991; Irwin & Zelinsky, 2002).

Perhaps the most compelling evidence for the presence of trans-saccadic information transfer and integration is that produced by analyses of fixations during reading. Examination of fixation durations and the correspondence between fixation location and the presented text has revealed the existence of extremely short fixations whose duration and positioning are determined by the

information obtained during preceding fixations (Morrison, 1984; Radach, et al., 2002). Given the unique nature of the task of reading (human-specific, highly practiced, characterized by organized scanpaths, etc.), the stability of these findings under other contexts remains an open question.

This paper presents a novel investigation of how humans actively view scenes of simple stimulus arrays when performing goal-directed visual search tasks, with two primary focuses: evidence for trans-saccadic information transfer and an examination of potential patterns in spatial attention.

Certain short-latency saccades (i.e. those preceded by short fixation durations) provide key insight into the process of trans-saccadic information transfer, as there is not sufficient time for visual information obtained during a short fixation to influence the saccade that follows; therefore, short-latency saccades must be executed based on other information sources: information carried over from fixations that occurred *before* the short-duration fixation.

There are several different hypothesized models of spatial attention, each admitting distinct predictions regarding the course of sequential saccades. One hypothesis posits that attention acts as a spotlight; observers tend to direct saccades towards the center of this local window of attention, within which neighboring stimuli also benefit from enhanced and more efficient visual processing. Under this type of facilitation mechanism, short-latency saccades directed to nearby stimuli would be more likely than those to farther locations. Conversely, there exists a model that proposes stimuli located near one's attentional fixation point are *suppressed*, leading to the claim that short-latency saccades to *distant* stimuli are more probable. Additionally, one may consider a form of spatial attention that incorporates both facilitation and suppression: nearby and distant stimuli fall within zones of facilitation, and thus serve as viable endpoint locations for

short-latency saccades, while medium-distance stimuli are highly unlikely to be the targets of short-latency saccades due to experiencing suppression.

2. Methods

2.1 Participants

Data was collected from 11 participants, recruited via advertisements on social media platforms. One subject withdrew due to discomfort with the task (physical strain on the eyes); the remaining 10 participants completed the entire course of the experiment. This cohort consisted of 6 men (average age of 23.7) and 4 women (average age of 26.5). There were 6 participants with normal vision and 4 who were near-sighted. A majority (3) of those who were near-sighted were able to engage in the behavioral tasks without the use of their eyeglasses, as indicated by their performance on a near visual acuity test. When text was displayed at a distance of ~36 cm, the participants who did not require eyeglasses were capable of accurately viewing text of 0.75 M or lower (which corresponds to better than 20/20), whereas the minimum text size that the participant who did require eyeglasses could see was 2.25 M.

2.2 Behavioral Tasks

Participants completed two goal-directed visual search tasks. One task involved a single target and a distractor, presented with a varying degree of separation (two-stimulus task). The other task varied the number of items in the stimulus array. For a given set size, the locations in which stimuli could appear were held constant (multiple-set-sizes task). These tasks are described in greater detail in sections 2.2.1 and 2.2.2 below. Use of two complementary manipulations supports the construction of a more robust, generalizable understanding of spatial attention. The two-stimulus task seeks to isolate the relationship between target-distractor separation and attentional selection; however, the systematic variation in said separation may induce strategic adaptations from participants. Pairing this task with the multiple-set-sizes task – which

introduces greater variation in task difficulty and the possibility of global inhibition (wherein focus on one stimulus may cause inhibition of all others) – helps elucidate whether the attentional mechanisms identified in the simple pairwise configuration extend to search scenarios with increased visual complexity. Additionally, inclusion of the multiple-set-sizes task is necessary to provide sufficient conditions for evaluating whether short-latency (second) saccades truly reflect integration of information from previous fixations.

On both tasks, participants were instructed to locate the target stimulus – defined as that with a “T” shape (Figure 1A) – in a stimulus array and indicate the direction of the horizontal bar’s shift relative to center with a corresponding key response of the left or right arrow key (Figure 1B).

Distractor items were those where the horizontal bar was *not* located at the top of the vertical bar; the horizontal and vertical bars could intersect at 7 possible (evenly spaced) positions along the vertical bar. For distractors, the horizontal bar was *not* shifted left or right relative to the center. The target and distractor(s) were each displayed at $4^\circ \times 2^\circ$ (width x height) of visual angle and were arranged in a circle with a radius of 12° of visual angle around the center of the screen.

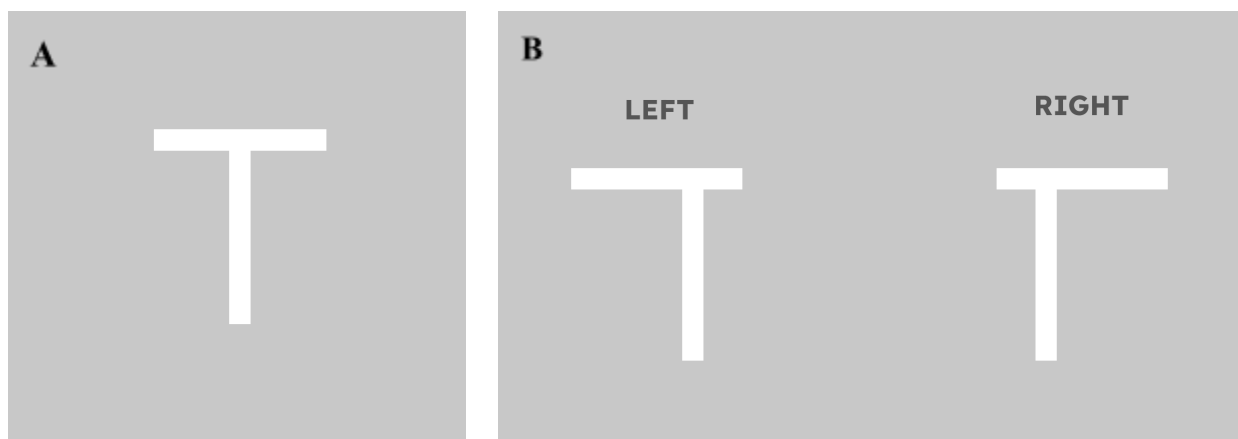


Figure 1: Visual aids shown to participants before commencing their first session to clarify the goal of the behavioral tasks. A) Target stimulus is that whose horizontal bar meets the vertical bar at the top of the vertical bar, forming a “T”-like shape. B) Two variations of the target

stimulus – left and right shifted. On each trial, participants must indicate which version is present with the appropriate key press.

2.2.1 Two-Stimulus Task

In the two-stimulus task, there were 16 evenly spaced locations along the circle at which the single target and the single distractor could appear. There were 8 conditions, corresponding to the number of separations (1 to 8) between the target and distractor. 1 separation indicates that the items were placed at two consecutive positions on the circle, while 8 separations reflect that they were directly across from each other (Figure 2A). The distractor was always defined as the same stimulus.

2.2.2 Multiple-Set-Sizes Task

This task involved 4 conditions: 2, 4, 8, and 16 stimuli. Unlike the two-stimulus task, distractors were not always of the same type. There were 7 distractor types (as was previously described) of varying similarity to the target. Under the conditions of 2 and 4 stimuli, random and distinct distractors were chosen. At 8 stimuli, there was exactly one of each distractor type, and for 16 stimuli, there were two of each distractor type as well as an additional, randomized one. For a given condition, the locations at which stimuli could appear were pre-defined; from among these options, the location of the target was randomized (Figure 2B).

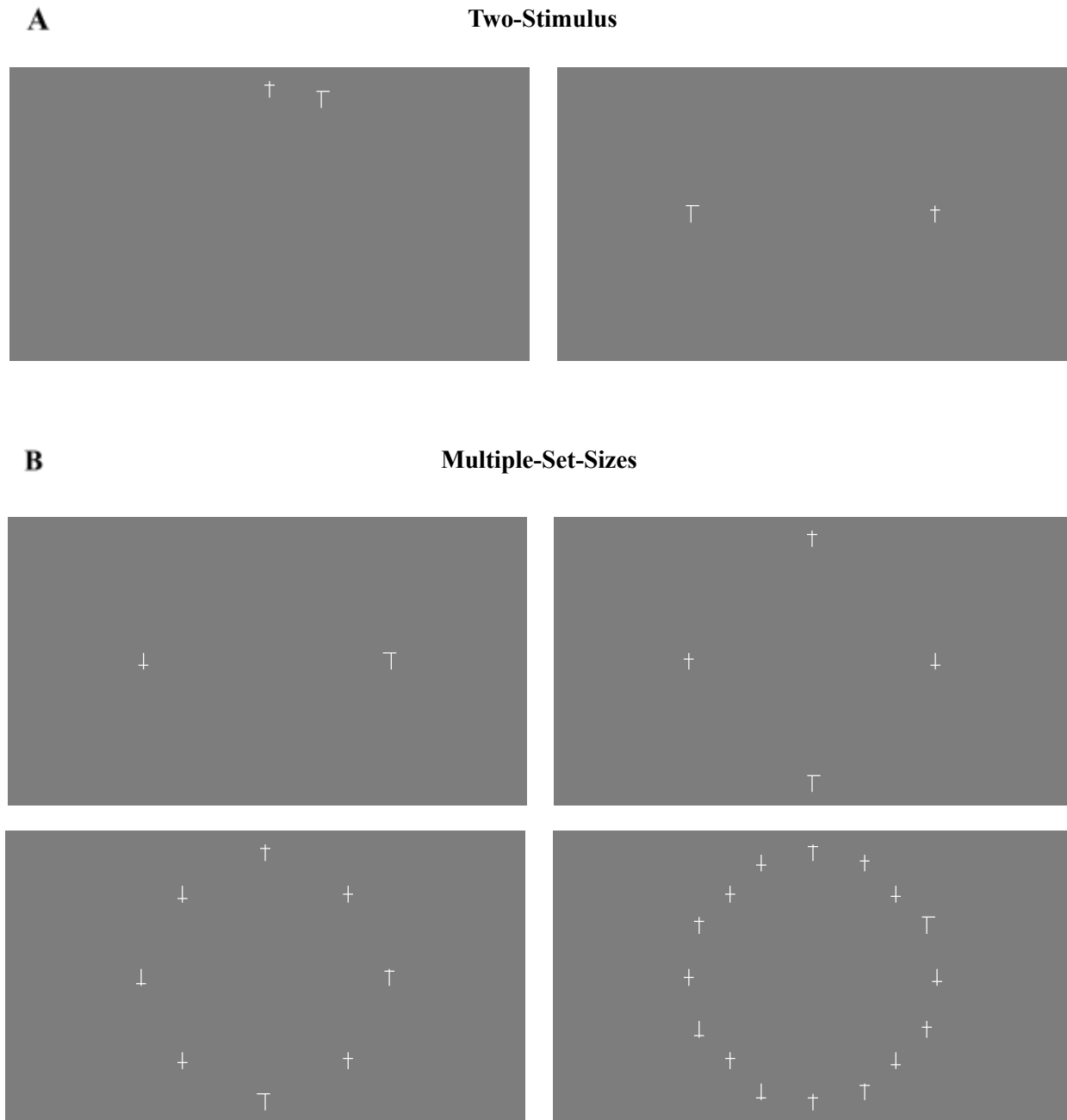


Figure 2: Examples of stimulus arrays for various conditions of both tasks. There were 16 positions at which the target and the distractor(s) could appear. These positions were evenly spaced around a circle with a radius of 12° of visual angle. A) During the two-stimulus task, there were no restrictions placed on which locations the single target and single distractor could occupy. The items could thus be separated by 1 to 8 position-steps, resulting in 8 conditions. Potential configurations for the 1 separation (minimal separation, items in consecutive positions) and 8 separations (maximal separation, items directly across from each other) conditions are

shown. B) Each of the 4 set size conditions of the multiple-set-sizes task was assigned a pre-defined spatial layout, which ensured even spacing between all stimuli present.

2.3 Apparatus

Stimulus arrays were presented to participants on a 53 cm ViewSonic XG2431 monitor (1920 x 1080 pixels) at a viewing distance of 57 cm, enforced by chin and head rests. Eye position was tracked at 1000 Hz using an EyeLink 1000 eye tracker (produced by SR Research Ltd). The eye tracking calibration and validation procedures involved sequential fixations of 9 static targets positioned at the center of the screen (0°), 20.48° toward each of the four screen corners, and $\pm 18.03^\circ$ horizontally and $\pm 10.37^\circ$ vertically from the center.

A photodiode attached to the upper left corner of the monitor was used to register the exact timing of stimulus presentation via a white rectangle positioned directly beneath the photodiode; the rectangle's onset and offset were synchronized with those of the stimulus array, admitting precise identification of the period during which the array was presented to the participant on each trial. This information was read and processed through a Teensy 4.1 device.

2.4 Experimental Procedure

Participants completed two sessions on two consecutive days, with the exception of one participant whose second session took place two days after the first. Prior to the primary behavioral tasks of the first session, tests were administered to assess distance and near visual acuity via Snellen and Jaeger eye charts. Both sessions consisted of 4 blocks with task alternation. If we code the tasks as A = two-stimulus and B = multiple-set-sizes, the orders of the first and second sessions' blocks were ABAB and BABA, respectively. Before the first two blocks of each session, participants were given a short set of practice trials (1 trial per task condition) corresponding to the upcoming behavioral task and intended to clarify and/or remind

them of the task's structure and goal. During this time, participants received feedback as to whether their responses were correct or not. They did not get feedback during the actual experiment.

Every block commenced with eye tracking calibration, allowing participants to step away from the apparatus, re-position themselves as desired, etc., between blocks. Each two-stimulus block was composed of 9 subblocks, with each subblock consisting of 32 trials (8 conditions x 4 trials per condition), while each multiple-set-sizes block contained 6 subblocks of 24 trials (4 conditions x 6 trials per condition). Subblocks were separated by optional minute-long breaks. No additional calibrations were necessary between subblocks since participants did not leave the apparatus during these breaks. Within a given subblock, the order of conditions was randomized. Across the full block, there were an equal number of trials in which the target shifted left versus right; the sequence of left and right-shift trials was also randomized.

At the beginning of each trial, a point was presented at the center of the screen for 1.5 seconds; participants were instructed to fixate the point for its entire duration. The central fixation point was followed by a stimulus array, which lasted until the participant's response. After indicating their response (and thus terminating the current trial), participants self-initiated the following trial via a key press; inter-trial intervals were therefore variable, though usually quite brief.

2.5 Data Analysis

2.5.1 Pre-Processing

Experimental event times (e.g. presentation of stimuli, participant responses, etc.) and eye tracking data were precisely aligned with each other for each trial. This included incorporating timing information obtained from the photodiode to account for unavoidable discrepancies

between the course of the experiment as run on the laboratory equipment and the true appearance of the stimulus array on the participant's screen. Accurate timing alignment is of great importance given that subsequent analyses are temporally fine-grained, examining eye movements at the scale of milliseconds.

A velocity-threshold algorithm was implemented to detect and extract saccades from the eye tracking data. First, the data was smoothed by grouping each set of 30 consecutive samples (in this case, with a sampling rate of 1000 Hz, 30 samples capture a window of 30 ms) together. The velocity threshold was defined as a velocity change of 30° (of visual angle) per second between two intervals. Saccade onsets were identified as the times when the velocity exceeded the threshold, and saccades terminated upon the velocity dropping below the threshold. Fixations were the periods between successive saccades. Additionally, two forms of filtering were enforced to achieve the final set of saccades: minimum duration of saccades, minimum duration of inter-saccadic fixation periods. Saccades with durations less than or equal to 4 ms were removed, and saccades that occurred within less than 2 ms of each other were merged.

The endpoint of each saccade was categorized based on which stimulus it was closest to, based on the Euclidean distances (in degrees of visual angle) between said endpoint and the position of each item in the present stimulus array. Participants were said to have foveated the target if their gaze was nearest to the target stimulus.

2.5.2 Visualization

Basic summary statistics regarding averages and percentages (Figures 4, 5, 6) of variables of interest were computed at each condition for each participant. For each condition, these values were averaged across participants.

Saccade latency distributions (Figures 3, 7) were generated by first calculating the proportion of saccades that fell into different latency bins – 0 ms to 400 ms with a step size of 20 ms – for each participant. These subject-level distributions were averaged together, taking the mean value for each bin across participants.

Group-level analyses that involved saccade latencies as the explanatory variable (Figures 8, 9, 10, 11) binned the latencies for each participant based on 8 equally spaced percentiles (12.5% per bin); the ranges of the bins dynamically adapted to the data of each specific subject. For each participant, the dependent variable of interest was calculated for the latencies that fell into each bin. For visualization purposes, the mean latency and mean dependent variable values were calculated across participants for each percentile bin.

2.5.2 Definition of Short-Latency

Short-latency saccades were defined as those that occurred after a fixation duration of 100 ms or less. 100 ms is an estimate of the minimum amount of time required to proceed from the onset of visual input upon a fixation to saccade execution – including visual processing, saccade goal selection, and saccade preparation. Thus, 100 ms corresponds to a critical threshold between saccades that can and cannot be influenced by the most recent visual input. This estimate was based on the latency distributions of first saccades from both of the tasks. Examination of these distributions revealed a clear separation of saccades into two populations around 100 ms (Figure 3): first saccades with short latencies that do not appear to be particularly stimulus-driven (reaching the target at only chance levels), and first saccades of regular latency that do.

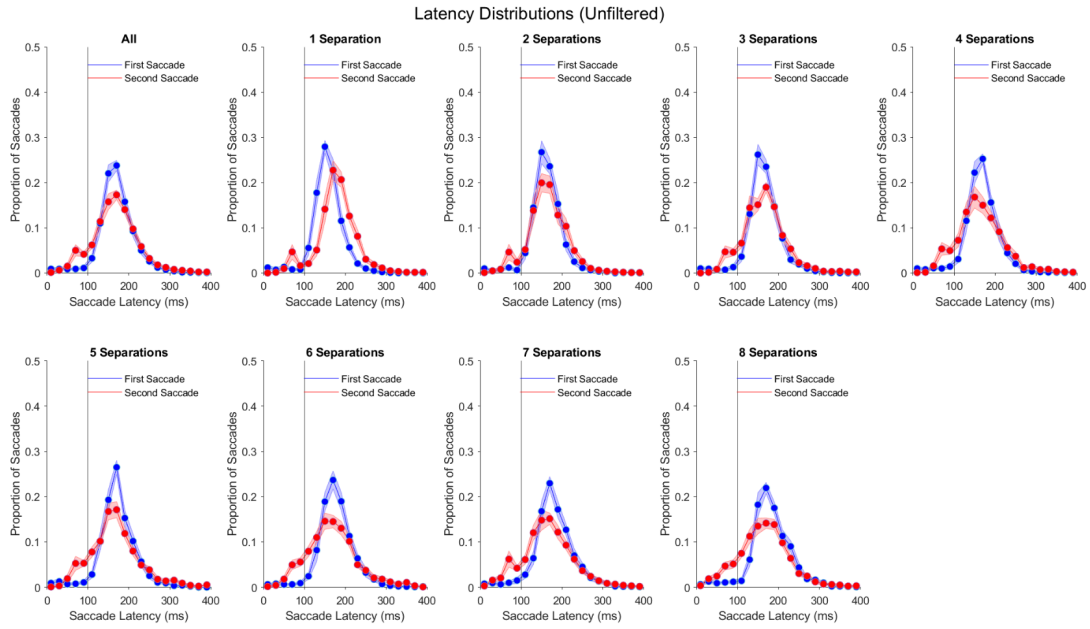
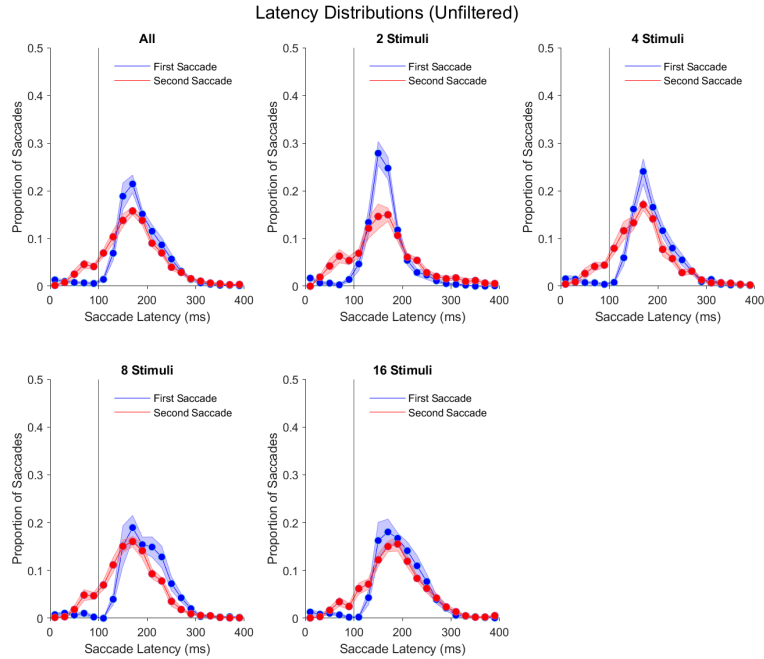
A**Two-Stimulus****B****Multiple-Set-Sizes**

Figure 3: Latency distributions of first and second saccades across conditions for both the two-stimulus (A) and multiple-set-sizes (B) tasks. Across tasks and conditions, there is an

apparent division of first saccades around 100 ms, with a steep increase in the frequency of first saccades above this boundary (100 ms is indicated on each graph with a vertical black line).

2.5.4 Excluded Data

Three blocks of data were removed from consideration during data analysis due to technical difficulties related to the equipment being used. These blocks came from 3 different participants, and were each for the multiple-set-sizes task.

Additionally, the reported findings exclude trials in which the first saccade occurred with short latency – defined above as any latency less than or equal to 100 ms. Short-latency first saccades were likely initiated before visual information about the stimulus array could be properly processed. Thus, they are unlikely to accurately reflect goal-directed perceptual evaluations and may instead be anticipatory in nature, divorced from the true stimuli at hand. Removal of these trials seeks to ensure that the retained data accurately capture participants' visual search and decision-making processes. Across all subjects, an average of 5.40% and 4.69% of all first saccades made during the two-stimulus and multiple-set-sizes tasks, respectively, were anticipatory.

3. Results

After the exclusion of certain data (as specified in section 2.5.4), analyses involved an average of 1089.80 ± 57.80 trials of the two-stimulus task and 508.20 ± 71.42 trials of the multiple-set-sizes task per participant. Key analyses regarding set-size effects observed during the multiple-set-sizes task as well as notable patterns of, and relationships between, first and second saccades from both tasks are reported below.

3.1 Set-Size Effects

Certain metrics of participant performance reveal expected effects of varying the size of the stimulus array to which they must attend. In particular, an increase in the number of stimuli presented corresponds with increases in the average number of saccades taken to locate the target stimulus and in average response time (Figure 4).

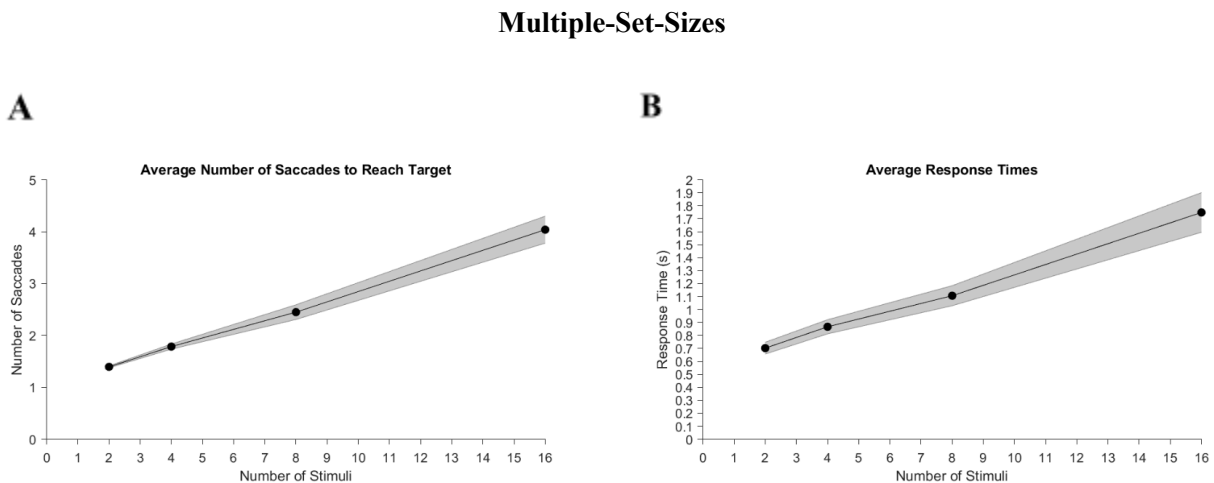


Figure 4: Basic set-size effects observed during the multiple-set-sizes task reflect expected changes in performance accompanying increased visual complexity of the stimulus array and thus increased task complexity. A) The average number of saccades taken to reach the target increases along with the set size. B) The average response time (measured as the time between stimulus array onset and the participant's key press) also steadily increases in relation to set size.

3.2 Saccade Latency Distributions

3.2.1 First Saccades

It appears that the average latency of first saccades increases both with a greater degree of separation between the target and a single distractor stimulus (Figure 5A) and as the number of stimuli increases (Figure 5B). Performance of a repeated measures ANOVA test on both tasks confirms that the average first saccade latency does indeed vary significantly across conditions. In the case of the two-stimulus task, $F(7, 63) = 51.956$ and $p < 0.001$ reveal a highly significant effect of stimulus separation on first saccade latency. Not only is this effect significant, it is also substantial, as a large effect size of $\eta^2 > 0.14$ indicates; a large proportion of the total variation in first saccade latency is explained by the spatial separation between the target and distractor. The multiple-set-sizes task also saw a highly significant and substantial effect (after Huynh-Feldt correction: $F(1.68, 15.10) = 28.332$, $p < 0.001$, $\eta^2 > 0.14$) – in this case, of set size on first saccade latency.

During the two-stimulus task, first saccade latency rises from 160.88 ± 3.59 ms (1 separation between the target and distractor) to 187.71 ± 3.92 ms (8 separations). A one-sample t-test ($t = 12.208$, $p < 0.001$) verified that there is a highly significant positive linear trend, with an (average) slope of +4.00 ms per additional separation ($SE = 0.328$, 95% CI [3.259, 4.741]). The multiple-set-sizes task sees an increase from 166.79 ± 4.00 ms (2 stimuli) to 198.92 ± 6.65 ms (16 stimuli) and carries a significant positive trend ($t = 5.199$, $p = 0.006$) of a +10.483 ms increase with each doubling of set size ($SE = 2.016$, 95% CI [5.921, 15.044]).

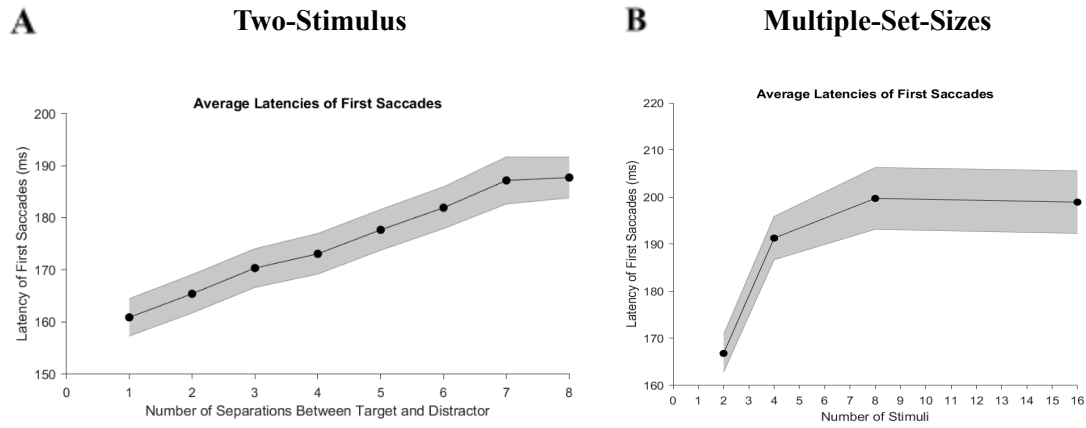


Figure 5: Repeated measures ANOVAs testing found highly significant and substantial effects of both stimulus separation and set size on first saccade latency. These effects can be captured by (significant) positive linear trends. A) With a high F-value, $p < 0.001$, and $\eta^2 > 0.14$ ($F(7, 63) = 51.956$, $p = 8.824 \times 10^{-24}$, $\eta^2 = 0.852$), the separation between the target and distractor has a highly significant and relatively large effect on first saccade latency. A one-sample t-test confirmed that there is a highly significant positive linear trend ($t = 12.208$, $p = 6.651 \times 10^{-7}$) of approximately +4.00 ms in first saccade latency per additional separation. B) There is a significant positive trend ($t = 5.199$, $p = 0.006$) of +10.483 ms in first saccade latency per doubling of set size that reflects the existence of a highly significant and substantial effect of set size ($F(1.68, 15.10) = 28.332$, $p = 1.435 \times 10^{-5}$, $\eta^2 = 0.759$).

3.2.2 Second Saccades

Unlike first saccades, the tasks display contrasting trends in the percentage of second saccades with short latencies. The likelihood of short-latency second saccades increases along with greater target-distractor separation, rising from $7.60 \pm 2.55\%$ at 1 separation to $18.43 \pm 3.08\%$ at 8 (Figure 6A) but decreases with a greater number of stimuli, dropping from $24.81 \pm 4.93\%$ with 2 stimuli to $8.04 \pm 1.72\%$ with 16 (Figure 6B). Overall, short-latency second saccades make up about $13.31\% (\pm 1.72\%)$ of all second saccades made during the two-stimulus task and $13.40\% (\pm 2.01\%)$ of those during the multiple-set-sizes task.

As in the case of the first saccades, the significance of these apparent trends were assessed via repeated measures ANOVA tests and one-sample t-tests. The percentage of second saccades with short latency is significantly affected by both stimulus separation (after Huynh-Feldt correction: $F(2.80, 25.23) = 3.224$, $p = 0.042$) and set size (after Huynh-Feldt correction: $F(1.38, 12.38) = 9.102$, $p = 0.007$). Both see large effect sizes: $\eta^2 = 0.602$ and $\eta^2 = 0.503$, respectively. However, the overall trends are in opposing directions; the two-stimulus task sees a significant positive trend ($t = 2.288$, $p = 0.048$) of +1.505 % per separation (SE = 0.658, 95% CI [0.017, 2.992]), while the multiple-set-sizes task reveals a significant *negative* trend ($t = -3.671$, $p = 0.005$) of -5.062 % per doubling of set size (SE = 1.379, 95% CI [-8.181, -1.942]).

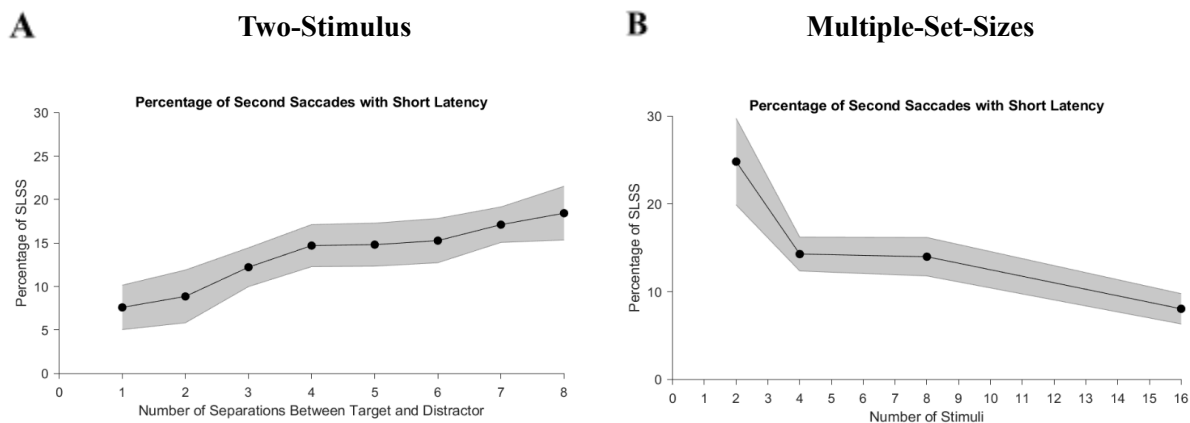


Figure 6: Repeated measures ANOVA testing confirmed that there are significant and substantial effects of both stimulus separation and set size on the percentage of second saccades that occur with short latency. The observed effects can be represented by trends of opposing directionality. A) Stimulus separation has a large, significant effect on the prevalence of short-latency second saccades, as evidenced by $F(2.80, 25.23) = 3.224$, $p = 0.042$, and $\eta^2 = 0.602$. This can be modeled through the significant positive trend ($t = 2.288$, $p = 0.048$) of +1.505 percent increase in short-latency second saccades per separation increase. B) There was also a highly significant effect due to set size ($F(1.38, 12.38) = 9.102$, $p = 0.007$, $\eta^2 = 0.503$), with a significant negative trend ($t = -3.671$, $p = 0.005$) of -5.062 % per set size doubling.

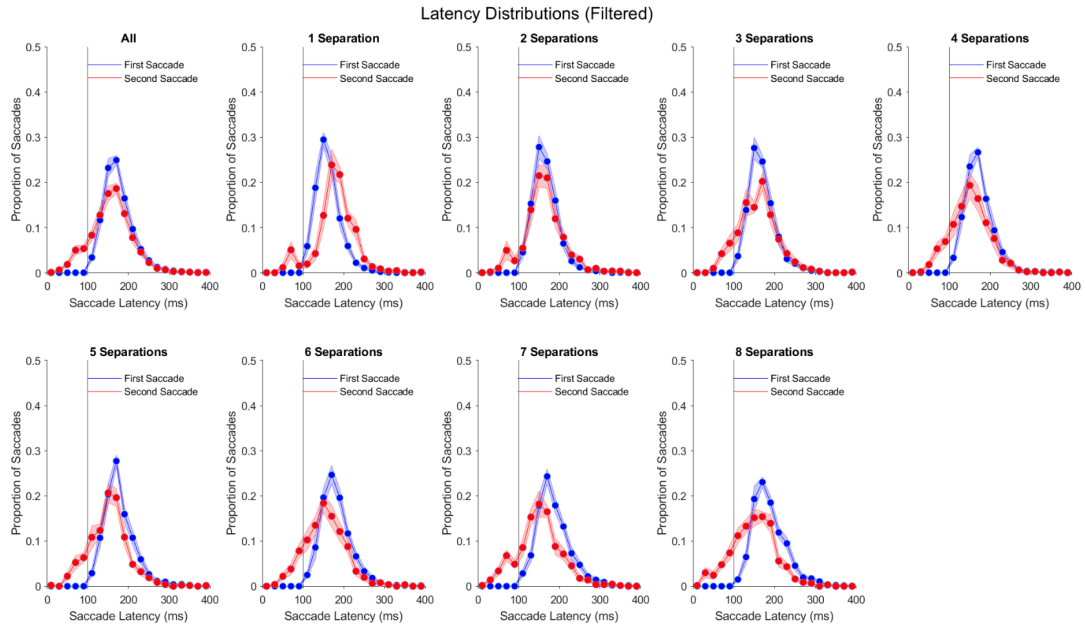
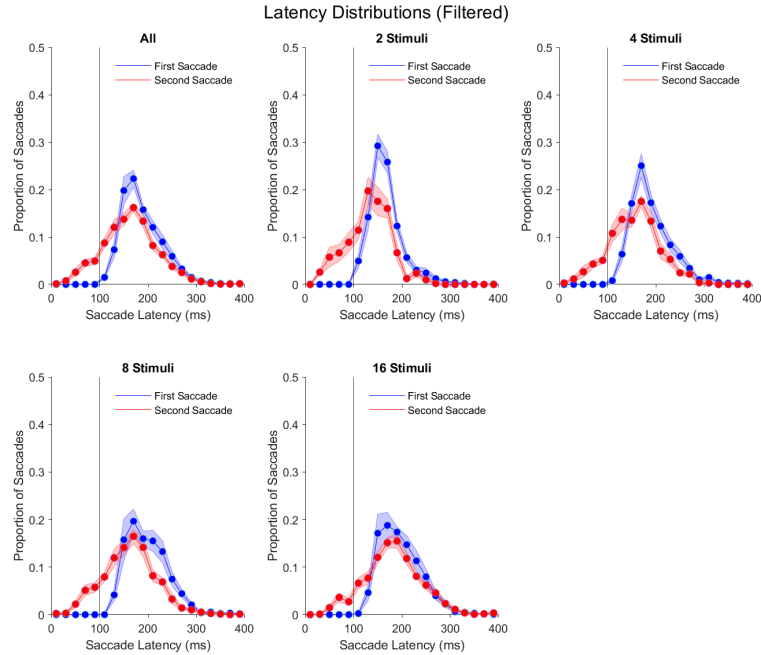
A**Two-Stimulus****B****Multiple-Set-Sizes**

Figure 7: Latency distributions of first and second saccades across conditions for both the two-stimulus (A) and multiple-set-sizes (B) tasks, excluding anticipatory first saccades that occur with latency ≤ 100 ms.

3.3 Second Saccade Latency in Relation to Preceding First Saccade Latency

The above findings regarding general trends in the latencies of saccades in relation to tasks' conditions are now extended to further probe the relationship *between* first and second saccades. For both tasks, pooling all conditions together leads to the observation that the likelihood of the second saccade of a trial occurring with a short latency increases as the latency of the preceding saccade (i.e. first saccade) increases (Figure 8A).

Within the two-stimulus task, Figure 8B highlights that this average can actually be decomposed into two groups which are stratified by condition and formed on the basis of whether there are meaningful gains in the probability of observing short-latency second saccades (as a function of first saccade latency) as the separation between the target and the distractor grows. Consider the change in short-latency second saccade probability seen across the first and last quantiles of the conditions of 2 separations versus 5 separations. At 2 separations, this probability oscillates in a range of $\sim 0.070\%$ to $\sim 0.125\%$, with $0.095 \pm 0.042 \%$ in the first quantile of first saccade latencies and $0.073 \pm 0.029 \%$ in the last. A positive trend is more apparent at 5 separations, where short-latency second saccade probability increases from $0.132 \pm 0.058 \%$ to $0.193 \pm 0.075 \%$. With a maximal separation between the target and the distractor (i.e. 8 separations), this trend is steeper; the probability of short-latency second saccades rises from $0.129 \pm 0.054 \%$ to $0.370 \pm 0.053 \%$.

This concept can be solidified via a linear mixed-effects model that contains random intercepts for participants and in which condition differences are incorporated as fixed effects. The model reveals minimal and non-significant variability in the probability of short-latency second saccades in relation to first saccade latency for stimulus separations up to 4, but a clear upward

trend emerges at separation 5, where β values (i.e. slopes) become both positive and statistically significant (Table 1). This pattern suggests that only at larger spatial separations may first-saccade latency reliably modulate the likelihood of short-latency second saccades.

Note that the above model was selected over that with random intercepts *and* random slopes for each participant based on a comparison AIC measures (random-intercept-only: 24061.52, both random: 24170.08) and log-likelihoods of the data under each model (random-intercept-only: -12013.8, both random: -12066.0). Selection of the current model over this alternative demonstrates that the observed effects are not significantly variable across different participants.

Stimulus Separation(s)	β	P-Value
1	+0.003	0.582
2	-0.002	0.693
3	+0.003	0.205
4	+0.007	0.062
5	+0.011	0.004
6	+0.014	1.7 e -5
7	+0.011	1.8 e -4
8	+0.009	0.002

Table 1: Slopes (β) and corresponding p-values of linear trends in the probability of short-latency second saccades versus first saccade latency for each condition of the two-stimulus task, determined via a linear mixed-effects model which allows random intercepts. There is no strong evidence for the presence of linear trends at stimulus separations below 5, while significant ($p < 0.01$ in all cases), positive trends are observed from 5 separations and higher.

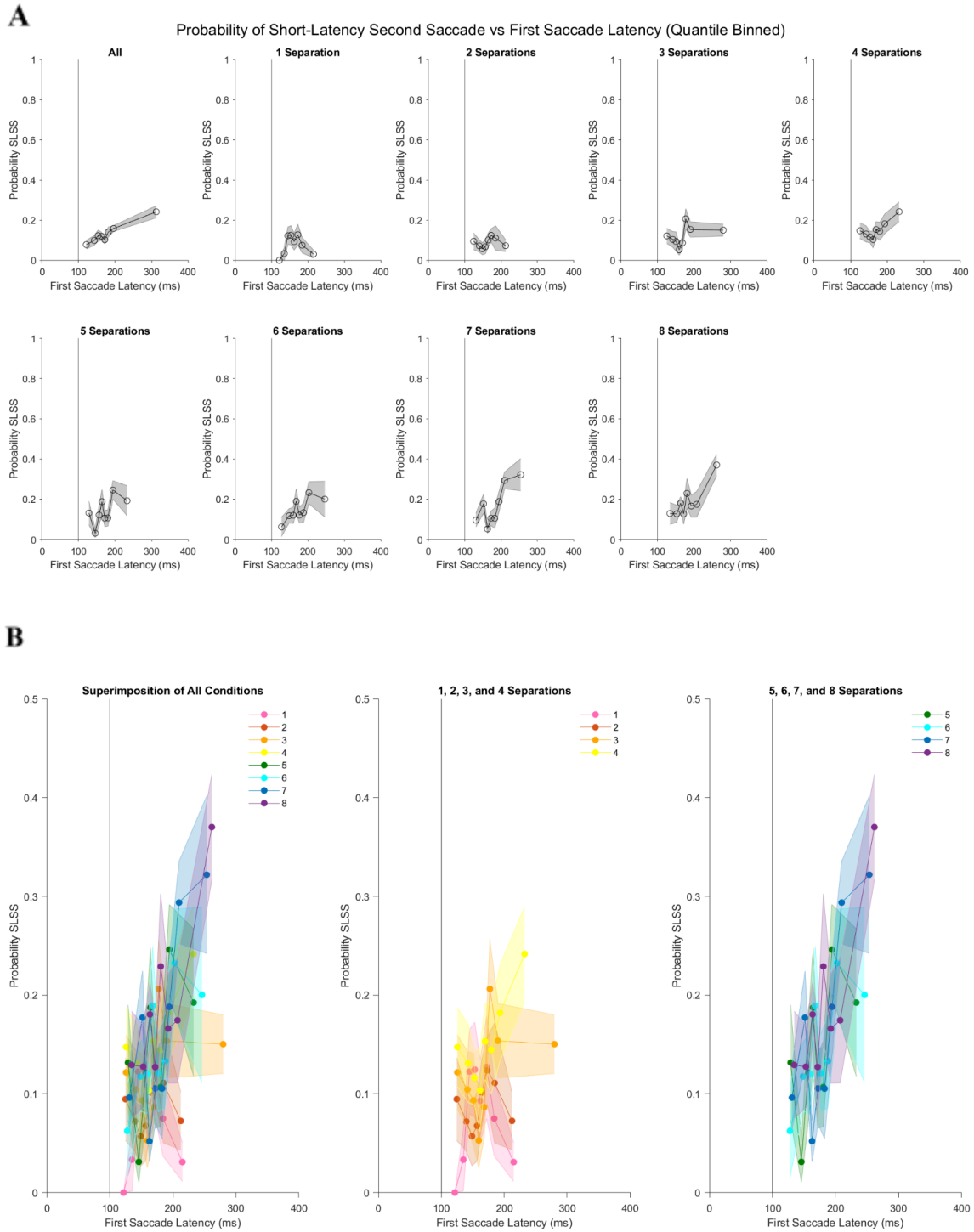


Figure 8: In the case of the two-stimulus task, two clear patterns emerge in the probability of second saccades occurring with short latency versus preceding first saccade latency. A) When pooling together trials across all conditions, it appears that, on average, short-latency second saccade likelihood increases with first saccade latency. However, there are notable

condition-specific variations around this average. B) A linear mixed-effects model with random intercepts found that at stimulus separations of 1 to 4 do there are *not* significant trends. At 5+ separations, significant, positive trends emerge. Here, conditions are grouped based on the significance (or lack thereof) of their trends (left: all, middle: $p > 0.05$, right: $p < 0.05$).

As previously stated, the multiple-set-sizes task reflects the same general trend as in the case of the two-stimulus task, and smaller set sizes tend to be associated with steeper slopes (Figure 9).

Again, for the sake of concrete illustration, compare the short-latency second saccade probability with respect to the first and last quartiles of first saccade latency across conditions: 2 stimuli: $(0.098 \pm 0.053 \%, 0.446 \pm 0.095 \%)$; 4 stimuli: $(0.049 \pm 0.026 \%, 0.221 \pm 0.036 \%)$; 8 stimuli: $(0.065 \pm 0.018 \%, 0.226 \pm 0.055 \%)$; 16 stimuli: $(0.021 \pm 0.014 \%, 0.166 \pm 0.043 \%)$. The significance of the observations is supported by a random-intercept-based linear mixed-effects model (AIC = 16614.68, log-likelihood = -8298.34; chosen over random intercept and random slope model with AIC = 16726.23, log-likelihood = -8352.12). Under each condition, there is a highly significant positive trend, and these slopes are generally of larger magnitude at smaller set sizes.

Set Size	β	P-Value
2	+0.020	2.5 e -6
4	+0.007	0.007
8	+0.007	7.5 e -4
16	+0.006	0.002

Table 2: Slopes (β) and corresponding p-values of linear trends in the probability of short-latency second saccades versus first saccade latency for each condition of the multiple-set-sizes task, determined via a linear mixed-effects model which allows random intercepts. All conditions display significant ($p < 0.01$ in all cases), positive trends. Smaller set sizes generally see trends

with steeper slopes: if A = given condition and B = condition whose set size is double that of A, then $\beta_A \geq \beta_B$.

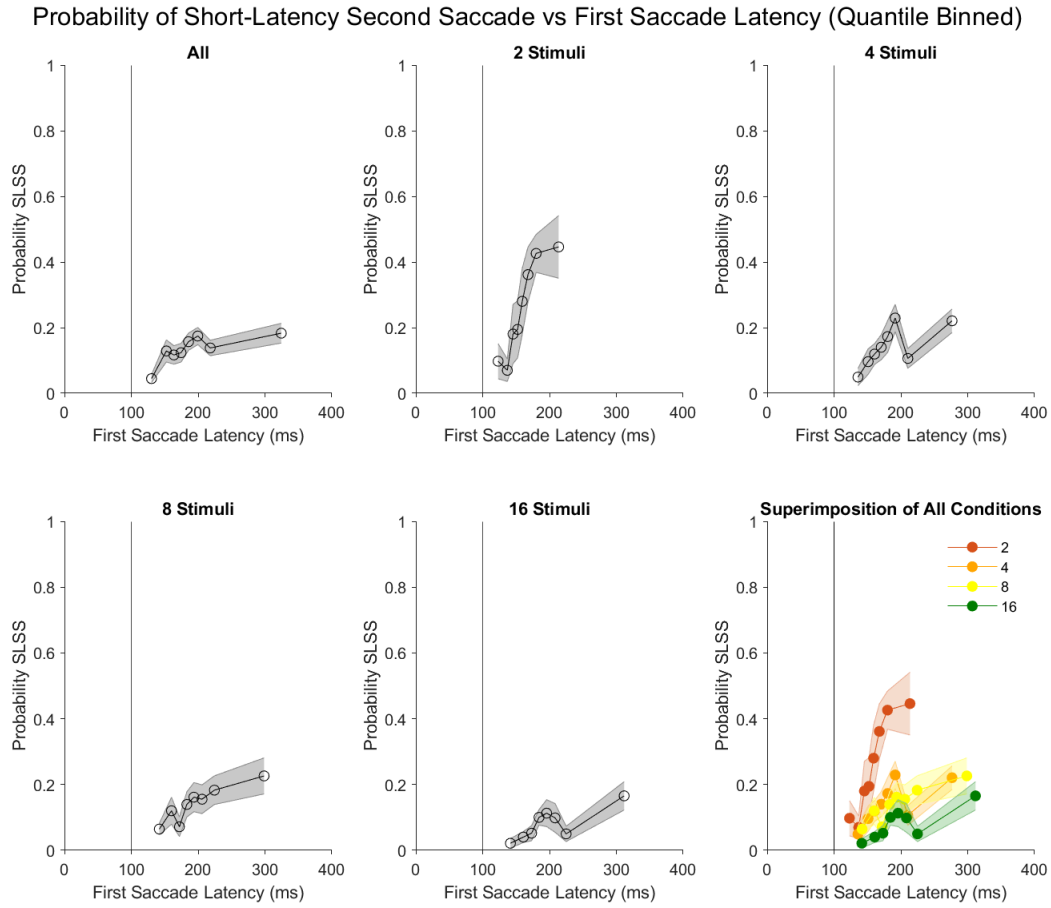
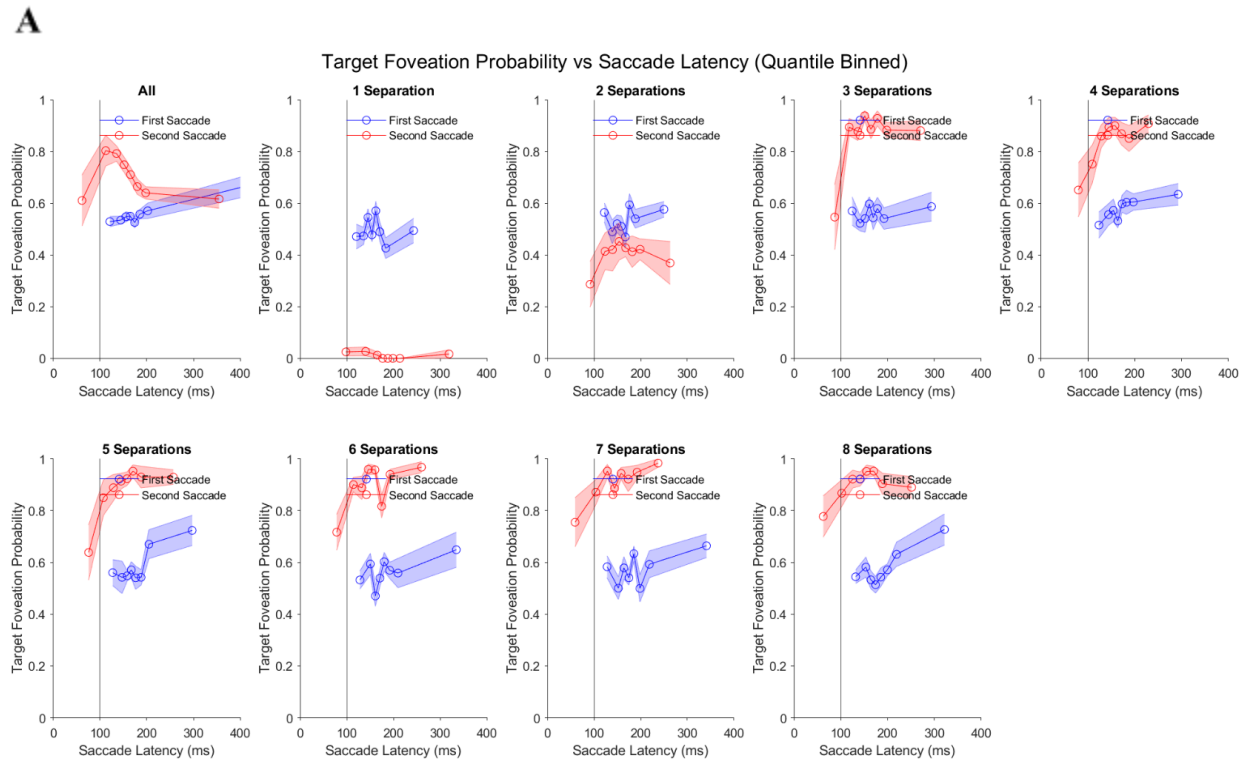


Figure 9: A linear mixed-effects model with random intercepts reveals that all conditions of the multiple-set-sizes task admit significant ($p < 0.01$) positive trends. In general, smaller set sizes are associated with steeper trends, as is visualized in the bottom-right plot.

3.4 Foveation of the Target in Relation to Saccade Latency

Analysis now focuses on the interplay between saccade latency and target foveation – namely, the probability of foveating the target stimulus as a function of first and second saccade latencies. A consistent (and expected) pattern across task and condition is an increase in target foveation probability on the first saccade as first saccade latency increases (Figure 10A).

On the two-stimulus task, it is also generally the case that target foveation probability on the second saccade increases with second saccade latency. Notably, second saccades foveate the target with relatively high probability relative to first saccades, including at short latencies. Of particular interest is the target foveation probability of short-latency second saccades, which increases as the stimulus separation increases (Figure 10B).



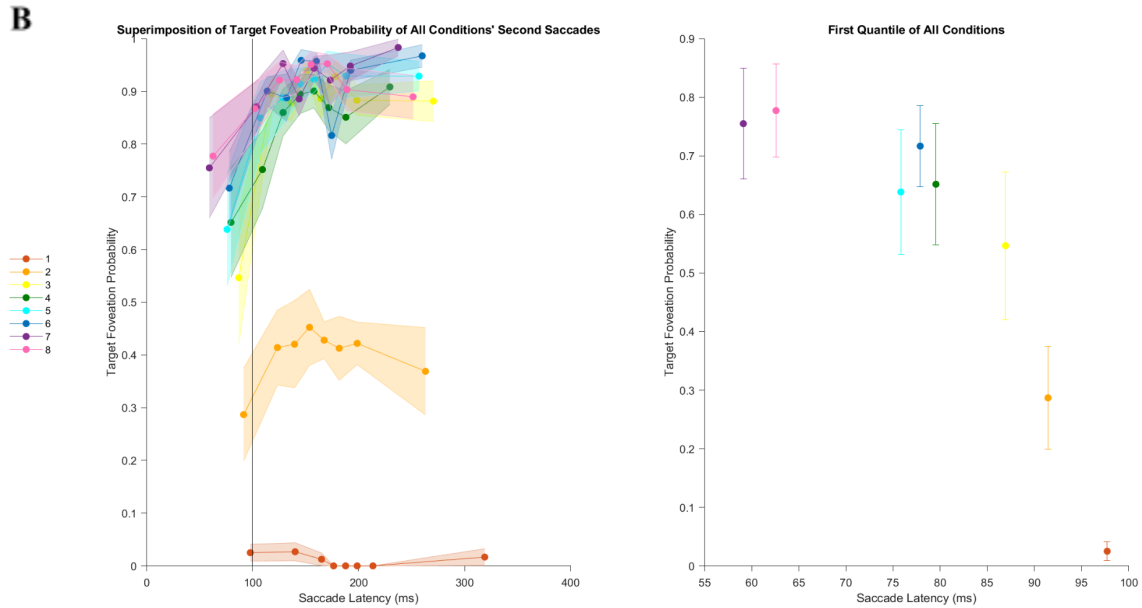


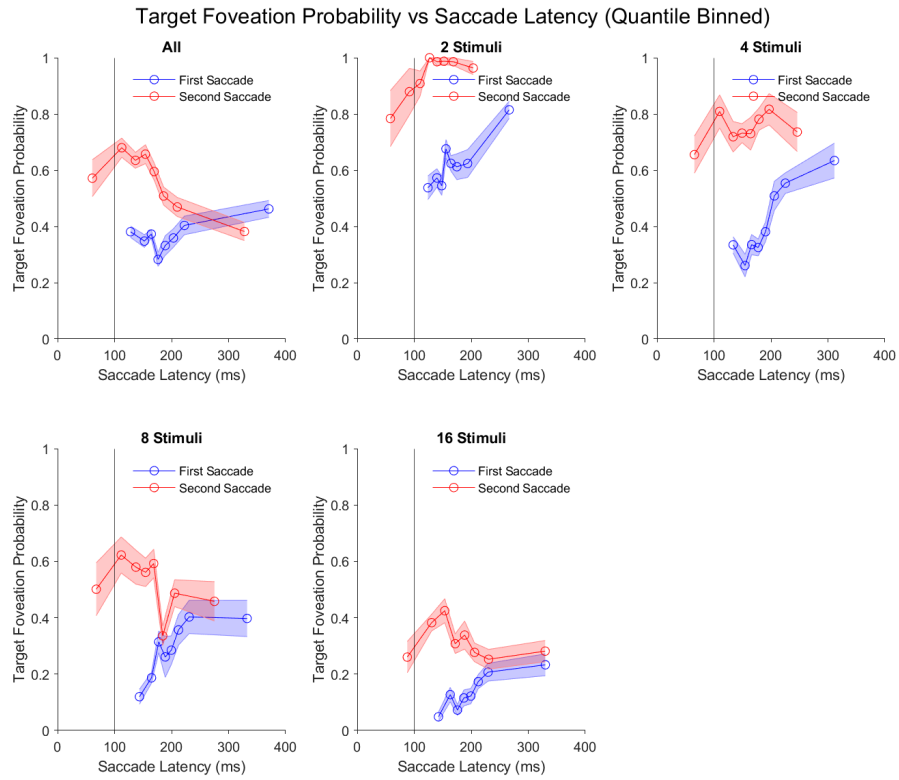
Figure 10: General patterns in target foveation probability versus latency of both first and second saccades are relatively stable across conditions of the two-stimulus task. A) Target foveation probability of first saccades reliably increases along with first saccade latency. A similar trend is seen with second saccades, where greater second saccade latency often (at stimulus separations of 3+) leads to a rapid increase in the likelihood of target foveation, followed by a plateau near 1. This is expected due to the nature of the task; if the target is not found on the first saccade, its identity and location must be the only other item in the stimulus array, allowing the second saccade to become a near error-free corrective movement. B) Short-latency second saccades appear to increasingly foveate the target as stimulus separation increases, with the most notable jumps occurring between the 1 and 2 separations conditions as well as 2 separations and 3+ separations.

A similar pattern is seen for the multiple-set-sizes task: target foveation probability is comparatively higher for second saccades than for first saccades, including short-latency second saccades. Target foveation probability in the event of short-latency second saccades decreases with larger set sizes; however, it consistently remains above chance (Figure 11). Given that a second saccade occurs (i.e. the participant did not locate the target at the preceding saccade) and assuming it does indeed terminate at a specific stimulus, the probability of locating the target by

chance is $\frac{1}{N-1}$, where N = set-size. With this in mind, it is notable that the target foveation probabilities of short-latency second saccades under the conditions of set sizes greater than 2 are as follows: 4 stimuli: 0.655 ± 0.066 % (chance = $0.\overline{3}$ %); 8 stimuli: 0.501 ± 0.094 % (chance = 0.143 %); 16 stimuli: 0.260 ± 0.055 % (chance = $0.0\overline{6}$) (Figure 11B). The 2-stimulus condition is omitted from consideration here due to the technical impossibility of assessing whether the observed target foveation probability is greater than that expected by random chance, since chance predicts a probability of 1.

This task does not reflect quite the same level of inter-condition pattern stability as the two-stimulus task. Under 2 stimuli, increased second saccade latency is associated with a steady increase and eventual plateau in target foveation probability. There is a less steady – though still present – increase in target foveation probability under 4 stimuli. Conversely, at 8 or 16 stimuli, increased latency of the second saccade does *not* reliably increase the likelihood of target foveation. Such set-size variability potentially serves as evidence that, for this visual search task, there exists some informational capacity at a set size between 4 and 8 stimuli. Around this inflection point, the visual system may become unable to maintain precise location information for all stimuli simultaneously, such that additional processing time (via increased duration of the initial fixation) cannot compensate for degraded or incomplete spatial encoding.

A



B

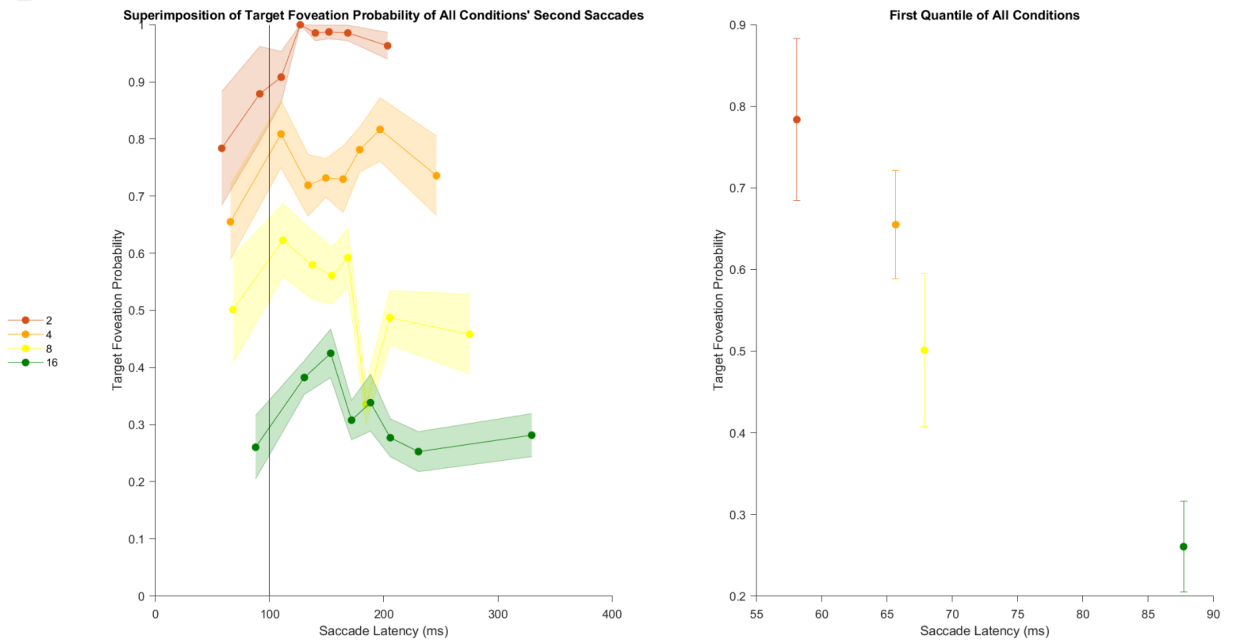


Figure 11: In the multiple-set-sizes task, trends in target foveation probability versus first saccade latency are stable across conditions but variable when considering second saccade latency.

A) Across all conditions of the multiple-set-sizes task, target foveation probability increases as first saccade latency increases. There is less condition invariance when it comes to examining patterns in target foveation in relation to second saccade latency. Notably, under large set sizes (8, 16), increased second saccade latency is not associated with an increased likelihood of target foveation (disparate from what is observed at smaller set sizes). This may be a reflection of an informational capacity limit around 4+ stimuli. B) Short-latency second saccades are capable of foveating the target with higher probability than chance (across all conditions for which such analysis is possible).

4. Discussion

The present study investigated trans-saccadic information transfer and spatial attention mechanisms during goal-directed visual search through an examination of saccade characteristics across two complementary tasks. Observed results provide compelling evidence that visual information can indeed be carried over across saccades and remain accessible to guide subsequent eye movements. Additionally, these results revealed patterns in saccades that are consistent with attentional surround suppression.

4.1 Evidence of Trans-Saccadic Information Transfer

During goal-directed visual search of stimulus arrays, first and second saccades were often separated by fixation periods of incredibly short durations (less than or equal to 100 ms). Across both tasks assessed, these short fixation durations made up about 13% of all second fixation periods. These durations do not provide sufficient time to collect information and meaningfully influence the saccades that immediately follow them (i.e. there is not enough time for the typical, full sequence of visual processing – including processing visual information, saccade preparation and execution – to occur). Therefore, short-latency second saccades must necessarily be based on visual information obtained earlier than their preceding fixations.

Supporting this claim, a general, task-invariant relationship was found wherein the probability that second saccades would occur with a short latency increased along with the latency of their preceding (first) saccades. Longer latency of the first saccade is equivalent to a longer initial fixation duration. During these extended fixation periods, participants are afforded more time to encode notable features of the stimulus array, and thus, more detailed information about the array is available prior to the first saccade taking place. Assuming the transfer of visual information across saccades, second saccades could thus be initiated rapidly, as the perceptual processing

which already took place during the first fixation reduces the amount of (and, in the case of quite simple tasks, may entirely eliminate the need for) new information that must be acquired via another fixation. The consistency of this pattern across both tasks, despite their differences in spatial geometry, complexity, etc., suggests that this finding reflects a relatively robust, fundamental property of visual processing and saccade planning during goal-directed searches (as opposed to displaying a task-specific strategy).

Consistent with this interpretation, short-latency second saccades are not randomly directed. Instead, they are clearly goal and target-directed, foveating the target with greater probability than chance (as analyzed through the multiple-set-sizes task) and occasionally even better than other, regular-latency second saccades. Such strong target selectivity implies that spatial information regarding potential targets is maintained across saccades. The visual system could not otherwise program a target-oriented short-latency second saccade; identification and localization of the target stimulus must be inherited from the initial fixation period.

4.2 Evidence of Surround Suppression in Spatial Attention

The positive trend in the percentage of observed short-latency second saccades across the conditions of the two-stimulus task provides compelling evidence for a mechanism of surround suppression in spatial attention. The prevalence of short-latency second saccades systematically increased with greater separation between the target and distractor. This suggests that when stimuli are close to each other, the selection of one location as the first saccade target actively suppresses neighboring locations, which, in turn, reduces one's ability to direct rapid subsequent saccades to said neighbor(s). As the separation between items increases, stimuli escape this inhibitory surround, allowing them to remain viable targets of subsequent short-latency second saccades.

The relationship between first saccade latency and the probability of a short-latency second saccade (in the context of the two-stimulus task) further supports this framing. Conditions with greater stimulus separation saw increasing short-latency second saccade probability with increased first saccade latency, whereas those with less separation did *not* reflect the same positive trend; this inter-condition variability implicates surround suppression. Greater separation between the target and distractor ensures that each stimulus lies outside the suppressed surround region associated with the other stimulus being selected as the attentional and saccadic target. In particular, when the first saccade is directed to the distractor (making later saccades necessary to locate the target), the target goes unsuppressed. The generation of a corrective short-latency second saccade to the target is thus possible, as sufficient information about the target could be processed during the initial fixation. In contrast, when stimulus separation is low, directing attention towards a given stimulus leads to suppression of the other, nearby stimulus. Regardless of the initial fixation duration, short-latency second saccades are rare due to the inhibited stimulus being too weakly represented to properly guide such saccades.

The multiple-set sizes task expands the view of surround suppression produced via the simple two-stimulus task by revealing the extent to which this mechanism scales in the face of stimulus arrays with increasing visual complexity. As the set size increased, there was a decrease in the percentage of second saccades that occurred with short latency. This may suggest that as the number of stimuli grows, local surround suppression transitions to a more distributed form of inhibition – i.e. the emergence of global inhibition, where focus on any given stimulus suppresses all others.

The relationship between the latency of first and second saccades during this task – with larger set sizes showing shallower positive trends in the probability of short-latency second saccades as

first saccade latency increases – introduces questions regarding the strength of global inhibition and the presence of a capacity limit to spatial memory. This reduction in short-latency second saccades could reflect: greater strength of inhibition exerted on stimuli, a breakdown in the retention of the formerly acquired spatial representations necessary to support short-latency saccades, as suggested by the capacity limit analysis, or some combination of the two.

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